

Semester Project:

Machine Learning and Predictive Analytics

BCIS 566

Raneem Ramadan

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Abstract

The purpose of this project is to apply machine learning techniques to a real-world dataset using Python. This project includes exploratory data analysis (EDA), preprocessing, model training, evaluation, and interpretation of results. Two machine learning models were also tested and compared.

The two models were the Artificial Neural Network (ANN) and the Random Forest as an bagging ensemble methods. The ANN presented an R-squared value of 0.477, and the Random Forest presented a value of 0.824. Therefore, the Random Forest presents better predictive performance. Additionally, evaluation metrics such as Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and Mean Absolute Error (MAE) were used for accuracy assessment.

The results that were presented show that income and geographic location are the most important predictors of housing price. This project shows how machine learning transforms data into meaningful analytics.

Introduction

Housing prices are influenced by a variety of factors, including but not limited to economy, geography, and demography. Predicting housing values is critical for policy makers, investors, and buyers. Machine learning is a tool that will assist in identifying these patterns to improve prediction accuracy. The goal of this project is to predict the median house value in California using machine learning techniques. Applying machine learning will allow the exploration of relationships between different variables and building predictive models. The two models that were compared, tested, and analyzed are the Artificial Neural Network and Random Forest.

Data & Material

The dataset is from the 1990 census data titled California Housing Prices. There is a total of 20,640 observations and over 9 predictor variables. The target variable (y) is `median_house_value`. This variable represents the median price of houses in each California

district. Since house prices are numeric and continuous, using regression models is the best option.

Variables Used

- longitude
- latitude
- housing_median_age
- total_rooms
- total_bedrooms
- population
- households
- median_income
- ocean_proximity (categorical variable)

Methodology

Data Preparation

The work was completed using Python libraries: numpy, pandas, tensorflow, and matplotlib.pyplot, seaborn, and scikit-learn.

After importing the data, it is critical to handle the missing values. Some missing values were found in total_bedrooms. To fix this, the missing values were replaced with the median to avoid errors and preserve the data.

```
dataset['total_bedrooms'] = dataset['total_bedrooms'].fillna(dataset['total_bedrooms'].median())
```

Another item to prepare the data is to transform the categorical data into numerical data. The ocean_proximity categorical variable was converted into dummy variables using the one-hot encoding method to help improve prediction accuracy, as it creates binary columns for each category. Where the specific category is set to 1 and the others to 0, like TRUE and FALSE. The output categories were as follows ...'ocean_proximity_INLAND', 'ocean_proximity_ISLAND', 'ocean_proximity_Less than 1 Hour Ocean', 'ocean_proximity_NEAR BAY', 'ocean_proximity_NEAR OCEAN. (from the end). This allows the model to present the

categories numerically. Converting these categories into numbers is necessary since machine learning models require numeric inputs, and most algorithms cannot process text.

```
dataset = pd.get_dummies(dataset, columns=['ocean_proximity'])  
  
print(dataset.columns)
```

In the original file, the title of the category was '<1H OCEAN'. This did not go through as a category due to readability concerns as part of the code, so it is preferred to replace the category name with 'Less than 1 Hour Ocean'.

```
dataset['ocean_proximity'] = dataset['ocean_proximity'].replace({  
  
    '<1H OCEAN': 'Less than 1 Hour Ocean'  
  
})
```

Train and Test Split

After preparing the data, the next step is to split the data into training set and testing set. With 80% training dataset and 20% testing dataset. This helps in validations.

Models Used

The models used are the Artificial Neural Network (ANN) and the Random Forest (RF). The ANN is a neural network that contains hidden layers. This model was selected to model complex nonlinear relationships. An RF is selected to combine many decision trees and perform on structured data.

Accuracy Assessment

The performance measures used were the R-squared, Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and the Mean Absolute Error (MAE). The R-squared value measures the percentage of variance in the dependent variable explained by the model, where values closer to 1 present better predictive performance. In addition to scatterplots.

The MSE measures the average squared difference between predicted and actual values- placing more focus on larger errors. The lower MSE indicates better model performance. The RMSE is the square root of the MSE and shows the error in original dollar units, making it more understandable in a real-world context.

Lastly, the MAE measures the average absolute difference between the predicted and actual values. Unlike the MSE, it is more interpretable as it does not focus on large errors. This presents better average prediction accuracy. Lower MAE shows better predictions.

Using a variety of metrics provides a more comprehensive assessment of model performance.

Results

Comparing ANN and Random Forest

The R-squared results from the ANN and Random Forest were compared to evaluate performance. The R-squared value for ANN is 0.477, while the R-squared value for Random Forest is 0.824. This indicates that the ANN explains approximately 47.7% of the price variation, while the Random Forest explains 82.4% of the price variation.

The significant improvement indicates that the Random Forest is a stronger model suited for the dataset. The higher R-squared presents a more accurate and reliable prediction compared to the ANN, which is likely due to its ability to analyze nonlinear interactions between variables in an effective structure. ANN is known for underperforming due to its limits and sensitivity to scaling structured tabular data.

Scatterplot

Scatterplots were used to compare the actual vs predictive housing values of the ANN and Random Forest models. When viewed closely, the plots help evaluate the model predictions against the actual data.

As shown in Figure 1, the ANN model displays wider spreads of datapoints, and many predictions are deviating from the diagonal line. The overall pattern is more of a horizontal line than a diagonal one, indicating a weak predictive performance with more error. Thus, the ANN

does not accurately predict the actual house price. Furthermore, the ANN requires more tuning as it is sensitive to feature scaling.

As for Figure 2, the Random Forest model shows more of a diagonal pattern with tighter clusters of points. This indicates better predictive accuracy as the values are closer to the actual values. The reduced spread of points also reflects low prediction error and high model accuracy. Overall, the ANN shows a wider spread, which means more error, while the Random Forest shows closer value points, indicating better, more consistent results.

Figure 1

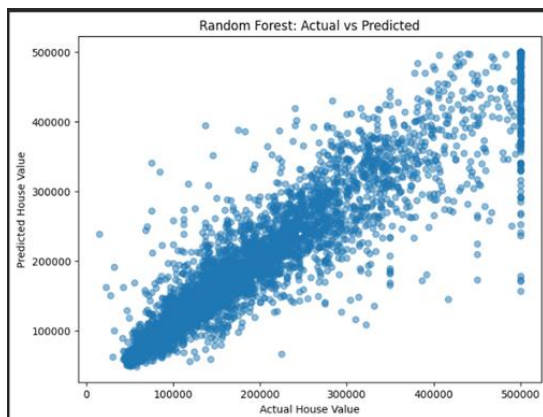


Figure 2

Both the visual and numerical results confirm that Random Forest outperforms the ANN model. Despite these results, Random Forest has some model limitations that can't be ignored. The model is computationally expensive and can not be tuned properly. Has a long runtime, specifically when measuring error using larger estimators.

ANN Training Error

The ANN training process was evaluated using the MSE learning curve to assess if the model was learning from the data. As seen in Figure 3, the model was decreasing over epochs, indicating that it was learning from the data. The validation MSE closely followed the training MSE, suggesting that the model did not overfit significantly. There was a minor gap in the error of the training and validation, presenting great generalization performance. Furthermore, the

curve gradually stabilizes in the final epochs. However, with the overall ANN model presenting a moderate R-squared of 0.477, the model is not suitable for structured data and cannot indicate appropriate predictive performance.

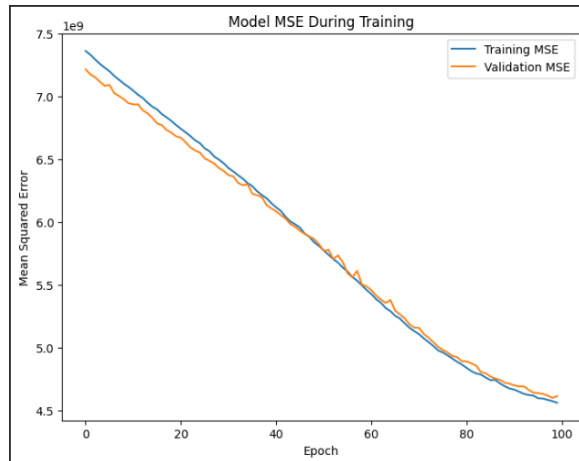


Figure 3

Error Metrics (Random Forest)

When measuring the error metrics, the RMSE for the Random Forest was approximately 47,914.77, indicating the magnitude of the prediction error in housing price units. The MAE value was approximately 31,414.97, which shows the average absolute difference between predicted and actual values. The MAE is lower than the RMSE, which indicates that the extreme errors are present, although limited. When testing different estimator values, the greater the number, the better the model performs. For example, a model with 400 trees has an R-squared of 0.825 (Figure 4), while the model with 200 trees has an R-squared of 0.8239 (Figure 5).

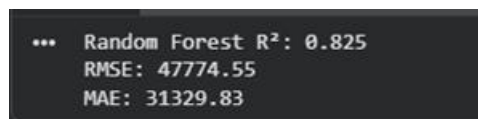


Figure 4

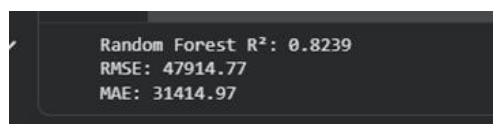


Figure 5

Correlation Heatmap

The correlation heatmap was created to measure the strength and relationship between numeric variables for the California Housing dataset. It helps determine which predictors are associated with house prices and supports feature selection. Using a heatmap allows for a better understanding of the variable relationships using colors, as it is faster to interpret the numbers.

The correlation heatmap analysis (Figure 6) shows that there is a strong positive relationship (0.69) between the median_house_value and those with high_income. Thereby, high-income households can buy houses at a higher price. As for the ocean_proximity_INLAND and the median house value, there is a moderate negative association (-0.48). This means that the farther a home is from the coast, the lower the price it is. All other variables in relation to median_house_value have weak to no correlation, indicating no relationship.

Some independent variables show relationships with one another and not with the dependent variables. Those are the total_rooms and total_bedrooms with a strong positive correlation (0.93). The household and total_bedrooms present the same idea that larger households have more bedrooms, with a strong positive correlation (0.97). Lastly, the population and households present a strong positive correlation with highly populated areas that have more people and householders (0.91).

Overall, the heatmap shows how the median house value has strong correlations with a specific variable, while other variables have moderate to almost no relationship to the price.

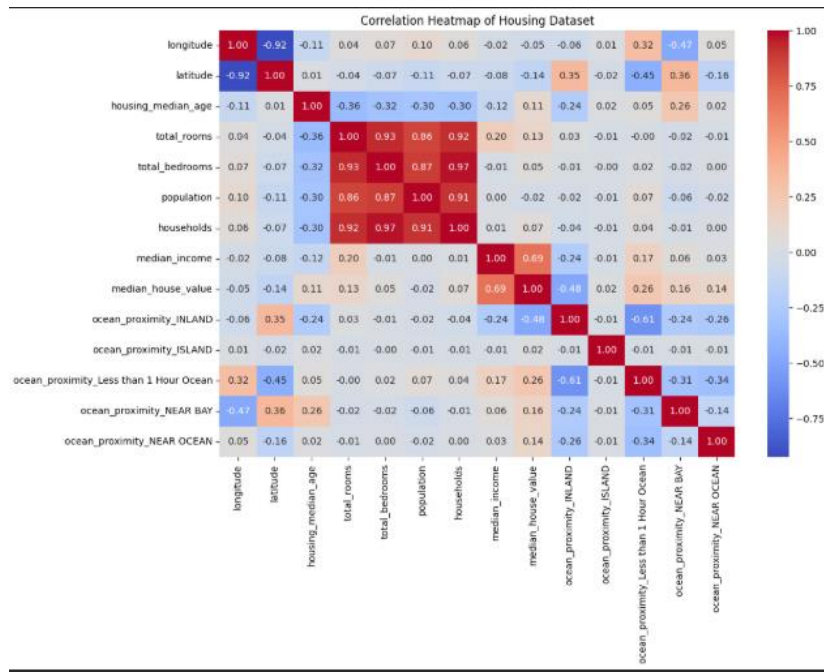


Figure 6

Feature Importance Analysis

The importance of features matters since it explains the variables that are most influential for predicting housing prices. It explains the model behavior and supports future feature selection. The top features explain how much that variable contributes to the model predictions.

Figure 7 shows that the median_income is the most important feature compared to other variables. Thus, it is the strongest predictor of housing prices in California. The second most important feature is the ocean_proximity_INLAND, which suggests that when the location is geographically inland, it impacts the housing prices more by decreasing the value. Furthermore, the longitude and latitude show moderate importance, which further confirms that the geographic location plays a role in determining housing prices. Other variables like the housing_median_age, population, total_rooms, and total_bedrooms have low importance scores. This means that they contribute less to predicting housing value compared to location and income. Overall, the feature importance aligns with the heatmap findings.

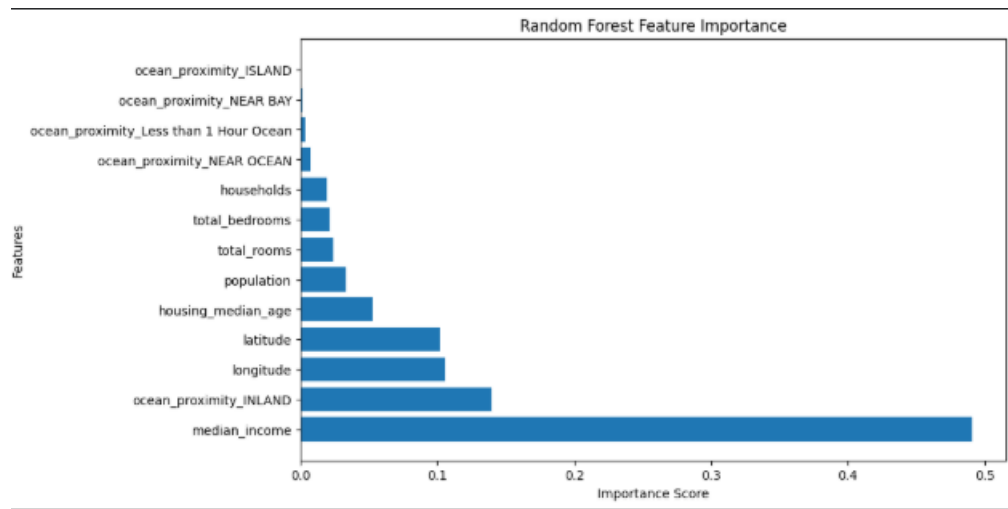


Figure 7

Discussion

The ANN was used to model nonlinear relationships within datasets since it is commonly used to work with regression models. The model uses ReLU functions in the hidden layer, the Adam optimizer is used for gradient updates, and the MSE is used for the loss function. The ANN shows a decreasing loss in epochs, which means that the model was learning from the data, yet the R-squared value of 0.4777 shows that there is only a moderate predictive performance. This indicates that the ANN is not a good choice for the structured tabular dataset. That being said, ANN has a limitation that requires hyperparameter tuning, which impacts performance.

In contrast, Random Forest was used to find an alternative approach to better capture nonlinear relationships and interactions of the variables. It constructs an ensemble of decision trees since it uses bootstrap and aggregation in their predictions to capture nonlinear relationships in a tabular data. This model also handles feature interactions better than an ANN. In the analysis, both 200 trees and 400 trees were used as estimators. The greater the number, the better the performance. However, it costs computational time. Overall, the Random Forest outperforms the ANN, especially since the R-squared value is 0.82, presenting better predictions.

ANN struggles despite learning the pattern during the training. The results conclude that for structured housing data, using a Random Forest is more effective in providing reliable results.

Recommendations and Improvements

In the future, it's suggested to use hyperparameter tuning methods such as Random Search to optimize model settings like max depth, learning rate, batch size, and neural network layers. Another recommendation would be to test ensemble methods like the XGBoost or model stacking to compare with ANN and Random Forest performance. In the future, one can also apply feature engineering by creating a new variable like population density. As for handling imbalances like outliers, using a KNN would be useful for missing value imputation. Using K-fold Cross Validation would be useful to reduce overfitting instead of a single train/test split. Collecting more recent housing data is another improvement for model predicting accuracy. Lastly, deploying this into Power BI can help the user receive real-time features and prices.

Conclusions

The project successfully applied machine learning techniques to analyze and predict housing prices in California. After developing the ANN and Random Forest (bagging ensemble methods), the results show that the Random Forest significantly outperformed the ANN and achieved an R^2 of 0.824. Furthermore, the strongest predictor of housing prices is the median income, followed by the geographic variables of ocean proximity and longitude and latitude. These were also consistent with the correlation heatmap and feature importance. Overall, the Random Forest model shows predictions that are more accurate as it can analyze nonlinear relationships. Future improvements include hyperparameter tuning, testing other models, and creating real-time applications like dashboards.

Reference

Géron, A. (n.d.). *Housing dataset* [Data set]. GitHub.

<https://github.com/ageron/handson-ml/blob/master/datasets/housing/README.md>

Nugent, C. (2017). *California housing prices* [Data set]. Kaggle.

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